

三、CSCA 化学考试大纲 (2025 版)

1. 考试目的 (Purpose of the Exam)

本考试旨在评估来华留学生攻读中国本科理工科专业所需的化学核心素养，包括化学知识体系的构建、科学思维方法的掌握和实验探究能力的培养。这些核心素养不仅是大学化学课程的基础，更是化学、化工、材料、环境等理工科专业学习的重要支撑，并为学生未来的科研创新与工程实践奠定坚实基础。

This exam aims to assess the core chemistry competencies required for international students to pursue undergraduate programs in science and engineering in China, including the establishment of a chemistry knowledge system, mastery of scientific thinking methods, and cultivation of experimental inquiry abilities. These core competencies are not only the foundation for university chemistry courses but also important support for studying science and engineering majors such as chemistry, chemical engineering, materials science, and environmental science, laying a solid foundation for students' future scientific research innovation and engineering practice.

2. 考试形式与试卷结构 (Exam Format and Paper Structure)

- ◆ 考试时长: 60 分钟 (Exam Duration: 60 minutes)
- ◆ 满分: 100 分 (Full Score: 100 points)
- ◆ 语言: 中文 / 英文 (Language: Chinese/English)
- ◆ 题型: 单项选择题 (Question Type: Multiple-choice questions (single answer))
- ◆ 题量: 共 48 题 (Number of Questions: 48 in total)
- ◆ 内容模块 (Content Modules):
 1. 化学基本概念及计算 (Basic Chemical Concepts and Calculations)
 2. 物质性质与反应 (Substance Properties and Reactions)
 3. 化学理论与规律 (Chemical Theories and Laws)
 4. 化学实验与应用 (Chemical Experiments and Applications)

3. 考试内容与范围 (Exam Content and Scope)

- ◆ 化学基本概念及计算 (Basic Chemical Concepts and Calculations)
 1. 物质分类与状态变化 (Substance classification and state changes)
 2. 化学用语与方程式书写 (Chemical terminology and chemical equation writing)
 3. 溶液浓度与 pH 计算 (Solution concentration and pH calculation)
 4. 物质的量相关计算 (Calculations related to the amount of substance)
 5. 理想气体状态方程应用 (Application of the Ideal Gas State Equation)
- ◆ 物质性质与反应 (Substance Properties and Reactions)
 1. 常见无机物性质 (单质、氧化物、酸、碱、盐) (Properties of common inorganic substances (elements, oxides, acids, bases, salts))
 2. 基础有机化合物 (烃类及衍生物) (Basic organic compounds (hydrocarbons and their derivatives))
 3. 氧化还原反应判断 (Judgment of redox reactions)
 4. 离子反应与检验方法 (Ionic reactions and detection methods)

- ◆ 化学理论与规律 (Chemical Theories and Laws)
 1. 原子结构与元素周期律 (Atomic structure and Periodic Law of Elements)
 2. 化学键与分子间作用力 (Chemical bonds and intermolecular forces)
 3. 化学反应速率与平衡 (Chemical reaction rate and equilibrium)
 4. 电解质溶液理论 (Electrolyte solution theory)
- ◆ 化学实验与应用 (Chemical Experiments and Applications)
 1. 实验室安全与仪器使用 (Laboratory safety and instrument use)
 2. 常见气体制备与检验 (Preparation and detection of common gases)
 3. 物质分离提纯方法 (Methods of substance separation and purification)
 4. 工业化工流程分析 (如合成氨) (Analysis of industrial chemical processes (e.g., ammonia synthesis))